Application of Machine Learning to Predict Groundwater Levels in Southeast Florida

MARTINA ROGERS*

Miami-Dade Water and Sewer Department, 3071 SW 38th Avenue, Miami, FL 33146

MICHAEL C. SUKOP
JAYANTHA OBEYSEKERA

Florida International University, 11200 SW 8th Street, OE 148, Miami, FL 33199

ANGELA MONTOYA
VIRGINIA M. WALSH

Miami-Dade Water and Sewer Department, 3071 SW 38th Avenue, Miami, FL 33146

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ABSTRACT

Miami-Dade County, FL, has complex hydrological and water management conditions, creating challenges for predicting groundwater levels. Due to the potential impacts of increasing sea level and changes in rain patterns on flooding and saltwater intrusion, it is essential to be able to accurately predict groundwater levels and to understand the drivers of groundwater levels. The method traditionally used to predict groundwater levels in south Florida and elsewhere is through the implementation of a numerical model. This research looks at machine learning techniques as an alternative method to provide improved predictions of groundwater levels. Extreme gradient boosting is one of the machine learning techniques that uses algorithms to train a model by identifying hidden patterns and relationships between groundwater level drivers and groundwater levels. The accuracies of the extreme gradient boosting model and a numerical model were compared by computing the root mean square error as a model performance measure. Comparing these two methods showed that the extreme gradient boosting model had better accuracy in predicting groundwater levels at the majority of groundwater monitoring sites. Furthermore, comparison of these two models helped us to identify problematic areas where both models performed poorly and narrow down the factors that could cause this poor model performance. Finally, the extreme gradient boosting method gave the importance of each feature included among the potential groundwater level drivers, including the ranked importance of a particular factor and its lag time, at each of the considered groundwater level monitoring sites.

INTRODUCTION

Miami-Dade County, FL, has complex hydrological and water management conditions, which create challenges in understanding groundwater level changes in response to many factors such as rain, evapotranspiration, ocean water levels, water levels in wetlands and marshes of the Everglades, and management of urban canals. Understanding and managing the water table are necessary to protect urban areas from flooding, keep the ecosystem healthy, and prevent saltwater intrusion into groundwater used for municipal well water supply. In general, flooding is caused by sea-level rise, storm surge, high tides, and rain, and the associated risk can be elevated when some of these factors are combined in a single event, leading to what is known as compound flooding (Jane et al., 2020). The most vulnerable areas are low-elevation communities where transverse glades cut through the highest area of southeast Florida, known as the Atlantic Coastal Ridge (Sukop et al., 2018), and other areas where the water table is near the surface, which limits the capacity of unsaturated zone storage during rainfall events. It has been observed that flooding days have been increasing with the continuous increase in sea level (Obeysekera et al., 2011; Wdowinski et al., 2016; and Sweet et al., 2018). Despite efforts to prevent flooding in these low-lying areas by managing canal water levels to keep the water table low, there are simultaneous efforts to keep the water table high enough to prevent saltwater intrusion from moving inland and

*Corresponding author email: martina.rogers@miamidade.gov

contaminating municipal wells. Regional-scale saltwater intrusion is monitored by the U.S. Geological Survey (USGS) (Prinos et al., 2014; Prinos, 2019). Another area where there is an effort to increase the water table is within Everglades National Park (ENP) and the Everglades more broadly, where one of the world's largest ecosystem restoration programs is working to counter long-term regional drainage that has threatened Everglades ecosystems.

Significant effort to understand and predict water levels in Miami-Dade County by USGS and Miami-Dade County resulted in the numerical Urban Miami-Dade Model (UMD) (Hughes and White, 2016), which uses the Newton-Raphson MODFLOW formulation (MODFLOW NWT) (Niswonger et al., 2011) with two additional packages: Surface Water Routing (SWR1) (Hughes et al., 2012) and Saltwater Intrusion SWI2 (Bakker et al., 2013). This model is complex and considers canal and surface water interaction with the groundwater and the movement of the saltwater front. Currently, this model serves as the best available tool to understand the groundwater table in Miami-Dade County and has been used as a base for numerous other studies, such as the projection of the water table to the year 2069 (Obeysekera et al., 2019) and understanding the impact of surge barriers on groundwater levels and flow (Obeysekera et al., 2021), and the fate of septic system discharge (Valencia et al., 2024). Building numerical models requires an understanding of the hydrogeology in the area, making assumptions, and setting up boundary conditions and aquifer parameters. Because of errors caused by limitations of the model, Hughes and White (2016) recommended interpreting the simulation results at timescales longer than the model's daily time step.

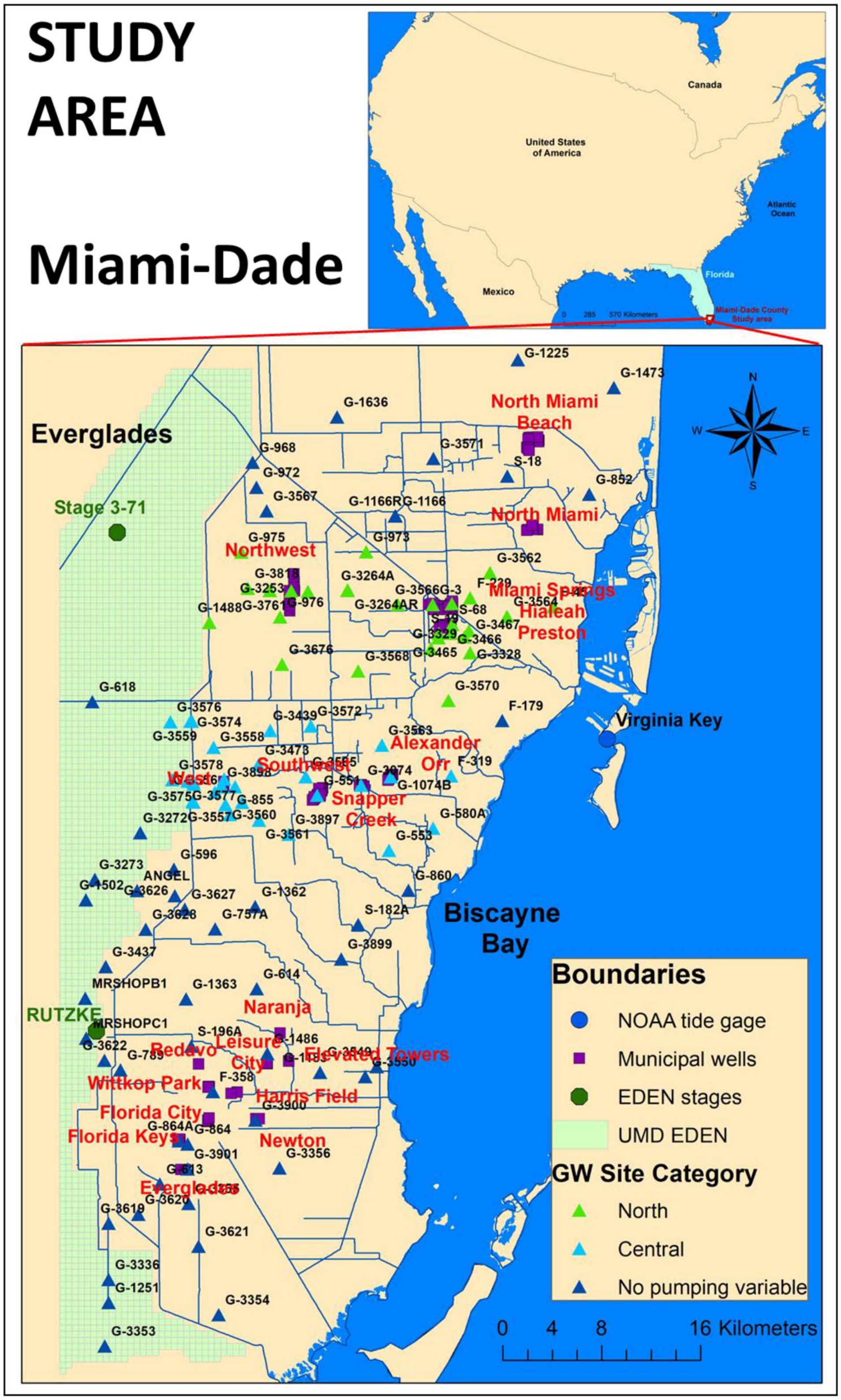
For a better understanding of key factors driving groundwater table changes and for shorter timescale (daily) result interpretation, data-driven and machine learning methods are becoming popular (Ahmadi et al., 2022; Osman et al., 2022; and Tao et al., 2022). Machine learning methods use regression-based modeling, are easier to use, and require only hydrological data; hydrogeological information is implicit in the observed relationships between potential drivers and the groundwater response. The ready availability of excellent machine learning algorithms enables data assimilation, produces accurate results, and has driven the popularity of these models in many areas of groundwater research in recent years (Di Salvo, 2022). Recent studies have continued this trend—for example, Xiong et al. (2023) demonstrated that boosting-based ensemble models such as XGBoost can achieve high accuracy for spatial groundwater predictions even in complex mountainous terrains, highlighting the algorithm's robustness across diverse hydrologic settings.

The input variables for machine learning are crucial for the accuracy of the results (Mohapatra et al., 2021). The most common input variables for groundwater level changes are precipitation and temperature (Kenda et al., 2018; Banadkooki et al., 2020; and Mohapatra et al., 2021). Many research papers have compared different methods of machine learning and found that extreme gradient boosting created one of the most accurate predictions (Naghibi et al., 2020; Park and Kim, 2021; and Osman et al., 2021). More advanced variations continue to emerge; for instance, Razavi-Termeh et al. (2024) integrated XGBoost with bio-inspired optimization algorithms to improve groundwater-prone area mapping, demonstrating the model's flexibility and ability to benefit from hybrid frameworks. Recent studies also explored hybrid spatiotemporal approaches: For example, Wang et al. (2024) combined graph convolutional networks (GCN) with long short-term memory (LSTM) to dynamically predict groundwater levels at multiple wells, accounting for both spatial autocorrelation and temporal dependencies. Extreme gradient boosting, as implemented in Python's XGBoost package (Chen and Guestrin, 2016), was applied in this research. XGBoost is one of the machine learning techniques that does not require extensive hydrological, geological, or spatial properties for the model but rather is based on the linear and nonlinear relationship of the dependent (target) and independent variables (predictor). However, there is growing interest in integrating machine learning approaches with MODFLOW numerical models, where hydrological parameters, such as hydraulic conductivity, can be estimated. For example, Baalousha (2025) applied XGBoost to estimate hydraulic conductivities, supporting MODFLOW calibration processes and demonstrating that combining a numerical model with machine learning can substantially reduce computational effort while maintaining high accuracy.

In this study, we focused on: (1) creating a groundwater level prediction model with an XGBoost method, (2) comparing the errors of the UMD numerical model for the calibration period of 2005–2010 with those of the XGBoost method, and (3) identifying the variables that most affected monitoring well groundwater levels. Understanding the variables that most affect the groundwater table in each area can be helpful for future groundwater management. This paper is organized into the following main categories: (1) description of the study area, (2) description of the existing Miami-Dade County numerical model, (3) description of the XGBoost method including input variables and model parameters, (4) performance measure selection, and (5) results and discussion of the performance of XGBoost and accuracy comparison of the machine learning techniques with the numerical model.

STUDY AREA

Miami-Dade County is located in a tropical monsoon climate zone distinguished by dry winter and wet summer


STUDY AREA
Miami-Dade
Canada
United States of America
Atlantic Ocean
Mexico
Florida
Miami-Dade County Study area
0 285 570 Kilometers
Everglades
Stage 3-71
North Miami Beach
North Miami
Northwest
Miami Springs
Hialeah
Preston
Virginia Key
Alexander Orr
Southwest
West
Snapper Creek
Biscayne Bay
Naranja
Leisure City
Elevated Towers
Redavo
Wittkop Park
Harris Field
Florida City
Florida Keys
Newton
Everglades
RUTZKE
G-1225
G-1473
G-1636
G-968
G-972
G-3567
G-3571
S-18
G-852
G-1166R
G-1166
G-975
G-973
G-3562
G-3264A
G-3264AR
G-976
G-1488
G-3676
G-3568
G-3465
G-3328
G-3466
G-3467
G-3564
S-68
G-3570
F-179
G-618
G-3576
G-3574
G-3559
G-3558
G-3439
G-3572
G-3473
G-3578
F-319
G-1074B
G-855
G-3560
G-3557
G-3272
G-3897
G-3561
G-553
G-580A
G-596
G-3273
ANGEL
G-1502
G-3626
G-3627
G-1362
G-757A
G-860
S-182A
G-3628
G-3899
G-3437
MRSHOPB1
G-1363
G-614
MRSHOPC1
S-196A
G-1486
G-3622
G-789
G-3550
F-358
G-3900
G-864A
G-864
G-3901
G-3356
G-613
G-3619
G-3620
G-3621
G-3336
G-1251
G-3354
G-3353
N
S
E
W
Boundaries
NOAA tide gage
Municipal wells
EDEN stages
UMD EDEN
GW Site Category
North
Central
No pumping variable
0 4 8 16 Kilometers

seasons with annual precipitation of 1,573 mm between 1990 and 2014 (Mohammad Harmay and Choi, 2023). With approximately 2.7 million residents, Miami-Dade County is the most populous county in Florida. It is located in southeast Florida, surrounded by the Atlantic Ocean on the east, the Everglades on the west, Broward County on the north, and the Florida Keys on the south. The topography has a low relief, with the elevation ranging from sea level to approximately 6.7 m at the Atlantic Coastal Ridge, parallel to the shore. Miami-Dade County is underlain by a surficial aquifer system, including the upper layers of the unconfined shallow Biscayne Aquifer, which contains layers with higher and lower permeabilities. The upper hydrogeologic units of the Biscayne Aquifer consist primarily of very highly permeable limestone in the geological units of the Miami Limestone, Fort Thompson Formation, and parts of Key Largo Limestone. The lower hydrological units consist mostly of confining sandstone and limestone (Renken et al., 2005). The Biscayne Aquifer has a thickness ranging from zero on the Everglades side to 40 m on the ocean side. It is affected by tidal fluctuations near the coastline (Rogers et al., 2023) and has exceptionally high hydraulic conductivity that may manifest as non-Darcian flow (Sukop et al., 2013), which conventional models cannot capture. It is one of the world's most productive aquifers.

Miami-Dade County contains 20 wellfields, comprising 139 municipal wells (Figure 1), and more wellfields are planned for the future. The groundwater is highly connected to canals that are used to drain large areas, to deliver water from the Everglades to the aquifer, and, ultimately, to discharge water to Biscayne Bay. High canal/groundwater connectivity is due to the surficial expression of high-permeability Biscayne Aquifer layers. Gated spillways, gated culverts, and pumps have been built to regulate the water in the canals to prevent flooding and saltwater intrusion. Water levels in the Everglades are available from the Everglades Depth Estimation Network (EDEN) (Haider et al., 2020).

METHODS

Description of Existing Numerical Model

The Urban Miami Model (UMD) was developed by Hughes and White (2016) and is a fully calibrated model with a grid of 101 columns and 189 rows resulting in a total of 19,089 cells with a cell size of 500 × 500 m. There are 15,853 active cells, of which 11,335 are considered onshore and 4,518 are considered offshore. The model has three layers, and they are all within the Biscayne Aquifer. Model boundaries and forcing variables include rain, reference evapotranspiration (RET), septic system return flows, agricultural and recreational wells, ocean water levels, the water level in Everglades from EDEN (Haider et al., 2020), and north boundaries based on three groundwater monitoring wells and one canal stage (Figure 1). The Everglades water levels include both the Water Conservation Areas (WCAs) in the north and ENP in the south. Gridded water level data with a 400 × 400 m spatial resolution were obtained from EDEN, a USGS hydrologic monitoring and modeling system designed to provide continuous, spatially distributed estimates of water surface elevation across the Greater Everglades region. EDEN integrates data from approximately 275 water level gauges and applies surface interpolation methods to generate daily water level estimates across the region (Haider et al., 2020). For this study, EDEN-derived water level data were interpolated to the UMD numerical model grid using the nearest-neighbor method to ensure spatial consistency between datasets. Canal stages and flows are generally not part of the boundary conditions because they are also simulated with the SWR package based on gate operations. The warm-up period of the model is 1 year, from January 1, 1996, to December 31, 1996. The calibration period is 8 years, from January 1, 1997, to December 31, 2004, and the verification period is 6 years, from January 1, 2005, to December 31, 2010 (Hughes and White, 2016). The calibration was performed with the automated parameter estimation software (PEST) (Doherty et al., 2010).

Extreme Gradient Boosting

Machine learning is one of the sub-fields of artificial intelligence (AI). The term was defined in a book by Arthur Samuel in 1950, where he described machine learning as the ability of a computer to learn without being programmed (Bell, 2022). Instead of giving a computer precise instruction to follow, machine learning allows computers to learn the program themselves from patterns in historical data. In general, more available data give more accurate predictions (Rebala et al., 2019). There are many different algorithms used in groundwater machine learning, including artificial neural networks (ANNs) (Sahoo et al., 2017), radial basis

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Figure 1. Map of the United States of America and study area of Miami-Dade County, FL. Purple squares represent municipal wells, single blue dot represents National Oceanic and Atmospheric Administration (NOAA) observed ocean water levels at Virginia Key, green dots represent Everglades water level stages, and blue lines represent canals. Green mesh represents UMD model EDEN boundary conditions. Triangles represent groundwater monitoring sites divided into three categories. Green triangles are groundwater sites grouped with municipal wells in the Miami Springs, Hialeah, Preston, and Northwest wellfields. Light blue triangles are groundwater sites coupled with Alexander Orr, Snapper Creek, Southwest, and West municipal wellfields. Dark blue triangles represent groundwater sites with no coupled wellfield.

function network (RBF) (Chen et al., 2020), adaptive neuro-fuzzy inference system (ANFIS) (Kayhomayoon et al., 2022), time-lagged recurrent neural networks (TLRNs) (Sundararajan et al., 2021), extreme learning machine (ELM) (Adnan et al., 2023), Bayesian network (BN) (Yin et al., 2021), instance-based weighting (IBW) (Xu et al., 2014), support vector machine (SVM) (Mohapatra et al., 2021), decision trees (DTs) (Kanyama et al., 2020), random forest (RF) (Liu et al., 2022), and gradient-boosted regression trees (GBRTs) (Di Salvo, 2022). In this research, the extreme gradient boosting (XGBoost) method was selected.

Extreme gradient boosting (XGBoost) was developed by Chen and Guestrin (2016) to improve speed efficiency through algorithmic optimizations. It is a tree-based machine learning model that works well with nonlinear relationships between dependent and independent variables. It is appropriate to use for rapidly processing extensive data, and it became popular because of its high prediction accuracy.

Application of Machine Learning to Miami-Dade Groundwater Levels

In this study, the dependent variable is groundwater level, and the independent variables are rain, evapotranspiration, ocean water levels, and water level in the Everglades, and, for some groundwater sites paired with municipal wells, well pumping rates. The variables for rain and evapotranspiration are total (NEXRAD grid) rain and RET and are the same as the inputs to the UMD USGS model preprocessing tool, which makes adjustments for runoff and land-use–based crop coefficients. Everglades water levels are represented by two water stage stations, RUTZKE and 3-71 (Figure 1), and were downloaded from the DBHydro Browser (South Florida Water Management District, 2023). These two stages were selected because they are stages of the EDEN network and not part of the UMD groundwater model sites. Their records go back to before 1996, they have no missing data in the study period, and they are a good representation of Everglades water levels in the northern part of the study area (WCAs) and south Miami-Dade County (ENP). The remaining independent variables (ocean and Everglades water levels, and pumping rates for some sites) were extracted from the UMD model (Hughes and White, 2016) from January 1, 1996, until December 31, 2010, in daily time steps. Missing data in the independent variables were detected only in the tide data, specifically a 35 day continuous gap in 1997. All other missing values occurred only in the dependent variable groundwater level time series. For both training and testing periods, missing data were interpolated linearly.

All the monitoring well sites were split into three categories. The first category contained 28 groundwater monitoring sites paired with four Miami-Dade Water and Sewer Department (MDWASD) wellfields (45 production wells). There were five independent variables (rain, RET, ocean water level, and two stations for Everglades water levels) considered at up to 14 day lag times plus the pumping data for the 45 wells. Inclusion of 0 to 14 day lags for each variable led to a total of 750 variables. The wellfields included in the first category were Miami Springs, Hialeah, Preston, and Northwest. The second category contained 31 monitoring wells paired with four MDWASD wellfields (34 production wells) and up to 14 day lag times for a total of 585 variables. The last category contained 53 groundwater monitoring sites not paired with municipal wells (Figure 1). Here, the lag time for all variables was also up to 14 days, making 75 variables. The rest (Figure 1) of the MDWASD (Leisure City, Newton, Everglades, Elevated Tower) and non-MDWASD wellfield pumping data (North Miami Beach, North Miami, Narnia, Wittkop Park, Redavo, Harris Field, Florida City, and the Florida Keys) were recorded in monthly time steps and then distributed equally throughout the month to daily time steps in the UMD model. These municipal well pumping data were not included as input to the XGBoost model because they cannot contribute to sub-monthly variability, which was a variable of key interest here.

Despite intensive canal-aquifer interaction, the stages and flows were not included as input to XGBoost; thus, the XGBoost model had the same principal inputs as the UMD model. Canal operation is based primarily on climatic conditions and ocean water level variables, which were included in the model.

The time series were split into training (1996–2004) and test (2005–2010) datasets, identical to the UMD model calibration and verification periods, except that 1 year of warm-up period in the UMD model was included in training datasets in the machine learning model because machine learning does not require a warm-up period. The training dataset represents the period where the model learns the relationship between different variables. In the numerical model, this period is known as calibration, where the parameters are tuned for the best performance of the model. The test dataset represents a period where the model accuracy is tested, and the simulated model results are compared with the observed groundwater data through performance measures. In the numerical model, this period is called verification. Groundwater monitoring wells with data records that started after 2005 or ended before 2005 (i.e., those without data in the training or testing period) were dropped from the analysis.

The most common parameters for tree booster are learning rate, maximum depth, colsample_bytree, and n-estimator. The learning rate is a boosting step, indicating the size shrinkage step. This parameter ranges between 0 and 1, and with larger values, there is a higher possibility of overfitting data. However, a smaller learning rate value requires a larger number of boosting stages, indicated by the parameter n-estimator (number of trees). A larger value of the n-estimator can lead to better performance, but at the same time, the larger number increases model run time. The n-estimator value ranges between one and infinity. Another common variable is maximum depth, which represents how far the nodes are expanded in the tree. Maximum depth value ranges between one and infinity. When building the trees to make a prediction, XGBoost allows only a certain percentage of columns (features) to build the tree. The colsample_bytree parameter assigns this percentage of features used in the tree and ranges from 0 to 1. The input XGBoost parameters were optimized manually through an iterative "trial-and-error" procedure to achieve the lowest prediction error. The tuning focused on four key hyper-parameters: learning_rate, max_depth, colsample_bytree, and n_estimators. Initial values were selected based on commonly recommended defaults (learning_rate = 0.1, max_depth = 6, n_estimators = 100, colsample_bytree = 0.8), and each parameter was systematically adjusted in small increments while monitoring the root mean squared error (RMSE) on a validation dataset. The process was repeated until no further improvement in RMSE was observed. The final parameter set—learning_rate = 0.1, max_depth = 1, colsample_bytree = 0.5, and n_estimators = 300—produced the lowest validation RMSE and was adopted for model training.

Performance Measures

There are several ways to validate the accuracy of a model, but a common metric is the root mean squared error (RMSE). The benefits of choosing RMSE as a model performance measure are that the error is in the same scale unit as the predicted value, and that it can be defined as an average error between n observed (y_j) and predicted ($\hat{y}_j$) values (Eq. 1). Smaller values of RMSE indicate better accuracy of the model.

$$RMSE \approx \sqrt{\frac{1}{n}\sum_{J=1}^{n}(y_j - \hat{y}_j)^2}. \quad (1)$$

The RMSE used in the Hughes and White (2016) UMD model was based on the weekly average of groundwater level observation and simulation values. In this study, the observed and simulated groundwater levels were paired on a daily time step, and the UMD model result errors were also estimated on a daily time step.

Table 1. *Summary of RMSE distributions for all groundwater monitoring wells during training (calibration) and testing (verification) periods. Values indicate the percentage of wells within three RMSE ranges: <0.1 m, 0.1–0.2 m, and >0.2 m, for both the XGBoost and UMD models.*

RMSE (m) Range	UMD Calibration	UMD Verification	XGBoost Train	XGBoost Test
<0.1	14%	16%	86%	43%
0.1–0.2	64%	50%	8%	46%
0.2–0.3	14%	24%	2%	3%
>0.3	6%	17%	0%	4%

RESULTS AND DISCUSSION

During the testing period, the XGBoost model showed RMSE values below 0.1 m at 39 percent of wells, between 0.1 and 0.2 m at 53 percent of wells, and above 0.2 m at 8 percent of wells. In comparison, the UMD model exhibited RMSE below 0.1 m at 12 percent of wells, between 0.1 and 0.2 m at 47 percent of wells, and above 0.2 m at 41 percent of wells. Overall, the XGBoost model produced lower errors than the UMD model during verification. Detailed RMSE values for all wells are presented in Supplemental Material Table S1 and summarized in Table 1 and Figure 2. The groundwater sites with the greatest RMSE (poorest model performance) for both models (UMD and XGBoost) are located near the Miami Springs, Hialeah, Preston, and Northwest wellfields in the C6 basin, near the Alexander Orr, Southwest, and Snapper Creek wellfields, and near the West wellfield (Figure 3). These wellfields have substantially higher daily pumpage rates than other county wellfields (Figure 4). The groundwater sites with the smallest RMSE in the UMD model are near the North and Everglades North boundaries, and for XGBoost, the majority of the area had low RMSE except west-central Miami-Dade County. The groundwater monitoring sites where both models performed with RMSE smaller than 0.1 m are G-1636 (Figure 5a), G-3549, G-3619, and G-3353.

For XGBoost, four groundwater monitoring sites had RMSE greater than 0.3 m in the testing period. These wells were G-1074B, G-1368A, G-3253, and G-551 (Figures 3 and 5b), with the highest importance variables being pumping rate at the Alexander Orr wellfield, the Preston wellfield, and the Southwest wellfield, respectively. For the UMD model, 18 wells had RMSE higher than 0.3 m in the verification period. These wells were G-3253, G-551, G-3273, G-3074, G-1368A, F-239, G-1074B, G-3259A, G-3627, G-3264AR, G-789,

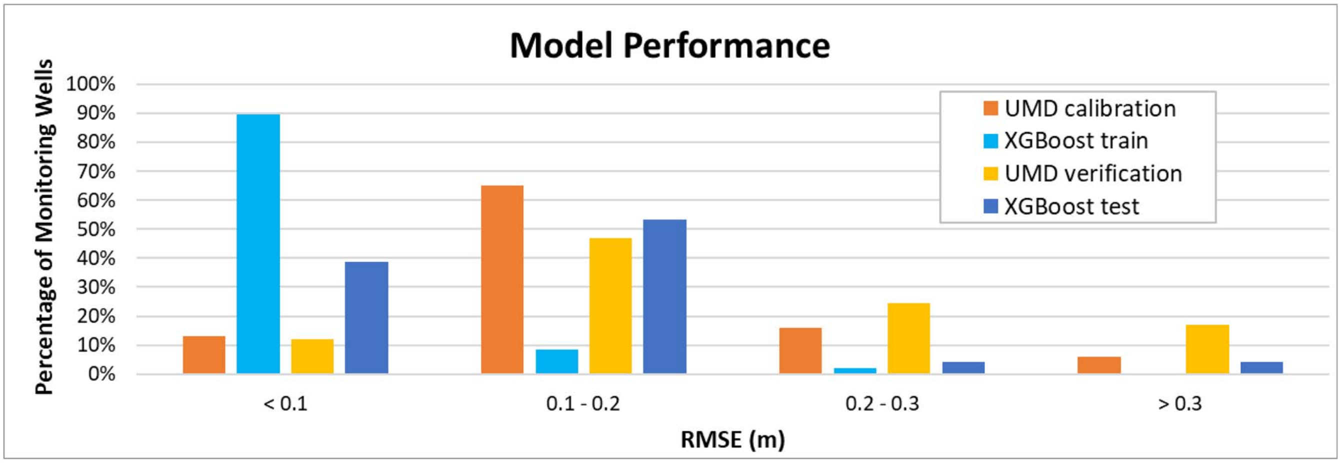

Figure 2. Model performance RMSE for the UMD and XGBoost models for training (calibration) and testing (verification) periods.

G-1363, G-3566, G-3564, G-3628, G-757A, G-3761, and G-3. However, there were other groundwater sites with the wellfield as the most important variable in the XGBoost model, and their RMSE was 0.11 m. These wells were G-3466 (Figure 5c) and G-3465. Overall, XGBoost showed higher accuracy for predicting groundwater levels near wellfields than UMD.

The distribution of the most important features across all groundwater sites is summarized in Figure 6. The pie chart illustrates the percentage of sites where each

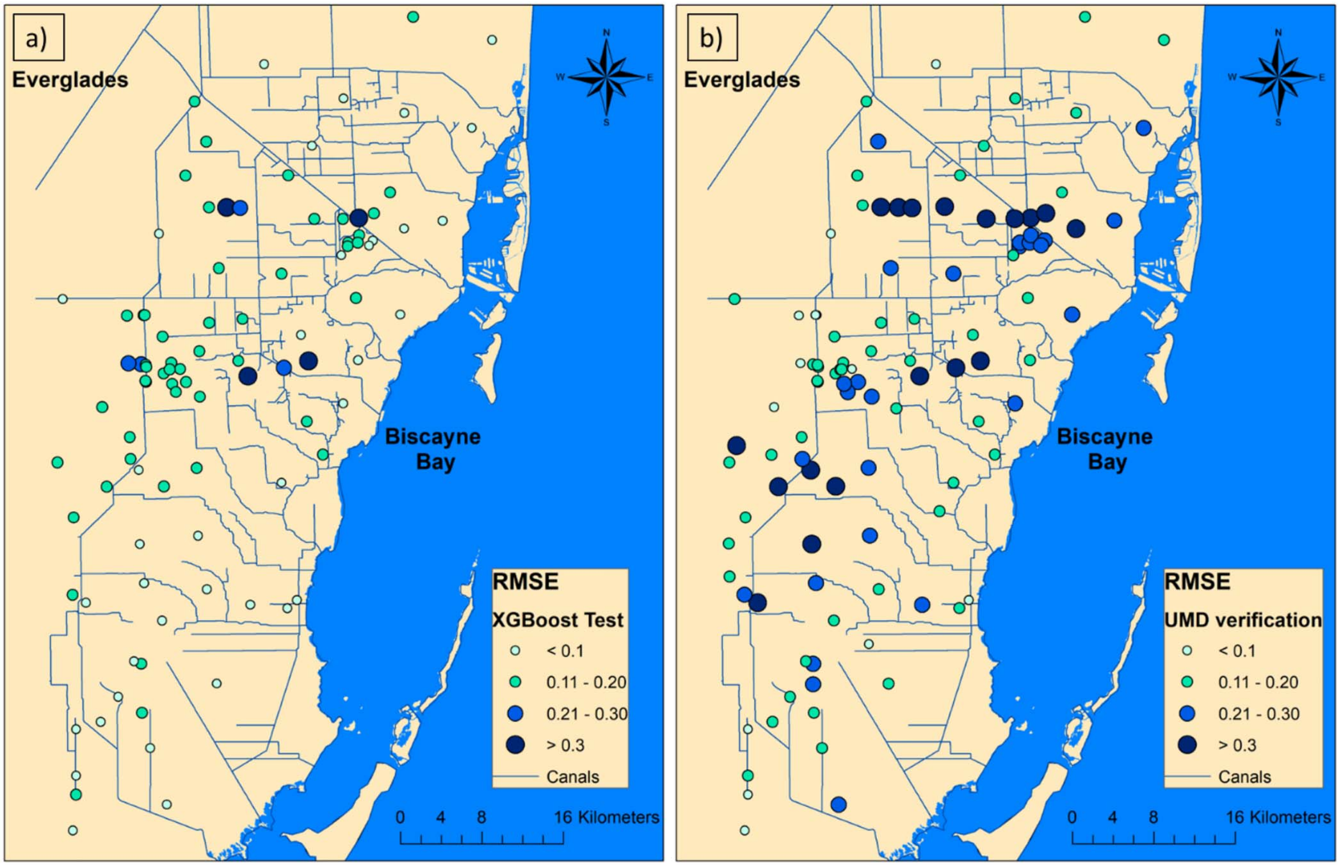

Figure 3. Map of Miami-Dade County groundwater sites with RMSE. Smaller RMSE (lighter blue colors) represents better model accuracy. (a) RMSE of the XGBoost model for the testing period. (b) RMSE of the UMD model for the verification period.

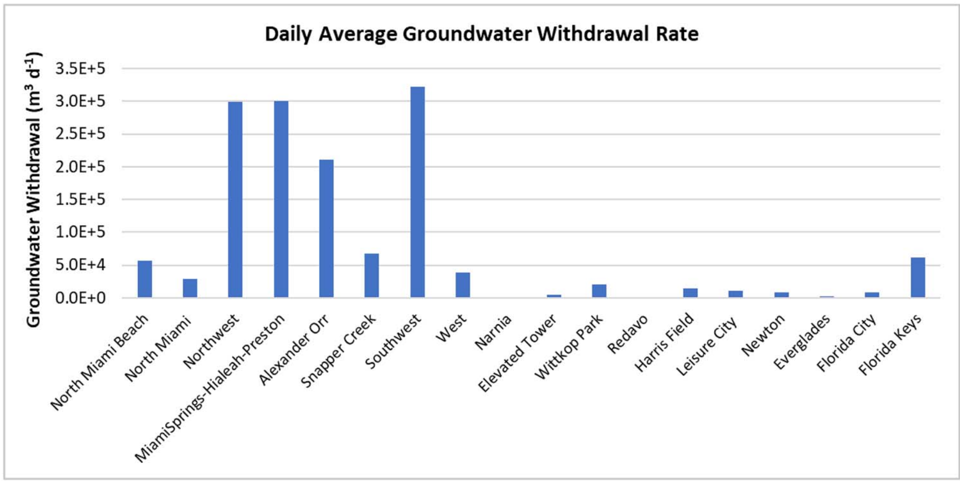

Figure 4. Daily average groundwater withdrawal rate for 20 wellfields. The Miami Springs, Hialeah, and Preston wellfields are joined because of their close proximity.

predictor, such as EDEN stages, pumping, or other variables, was identified as the most influential feature. The most-important-feature map (Figure 7) shows the variable affecting the groundwater site the most and its lag time in days. The most important considered variable for the majority of groundwater sites was the water level in the Everglades, illustrated on the map as EDEN (Figure 7). EDEN for the XGBoost model is represented by two EDEN stages, Rutzke and 3-71 (Figure 1). This result could be explained by the similar impact of rainfall on Everglades water levels and the water table elevations in much of the area. Only four groundwater sites (G-3327, G-3562, G-3467, F-319) had the ocean water level as their most important variable (Figures 6 and 7). Three of these four groundwater sites had a small RMSE of 0.07 m, and one had RMSE of 0.11 m. The attenuated tidal signal observed in the mainland wells aligns with findings from a prior study (Rogers et al., 2023). In that previous analysis, groundwater well time-series data were examined using cross-correlation functions, tidal efficiency measures, and the fast Fourier transform (FFT) method. Six groundwater sites had rain as the most important variable, and all of those had RMSE smaller than 0.12 m. Only one groundwater site, G-3549, had evapotranspiration as its most important variable, with RMSE of 0.08 (Figures 3 and 6). Figure 7 highlights the most important feature for the water level in each well, showing the variable and its lag time, but all the variables had some role in predicting the groundwater level. XGBoost fell short in predicting lower values of the water level time series (e.g., Figure 5a). The prediction of groundwater levels could likely be improved by having more values incorporated into the training dataset.

To assess whether the water levels in the Everglades could have a predominant influence on water levels at groundwater sites, as suggested by the machine learning model, numerical model simulations were conducted by elevating the boundary on the Everglades side by 0.6 m over the whole simulation period 1996–2010 as well as during a single-day spike event. The choice of a 0.6 m increase was based on observed water level trends in the eastern Everglades, where elevations have risen by approximately 0.3 m between 2010 and 2018 due to ongoing Comprehensive Everglades Restoration Plan (CERP) restoration activities, as determined from USGS EDEN stage data (Haider et al., 2020). Given the continued progress of restoration projects such as the Biscayne Bay and Southeastern Everglades Ecosystem Restoration (BBSEER) Project—which aim to increase freshwater connectivity and sheet flow (USACE, 2021; Harvey et al., 2024)—additional future increases in stage levels are plausible. Therefore, the 0.6 m scenario was selected to represent a reasonable and realistic increase beyond the observed change, allowing the model to explore potential impacts under likely future restoration conditions.

The impacts of these changes were assessed by subtracting the original heads in the model from the new heads, considering the raised boundaries both over the whole simulation period 1996–2010 (Figure 8a) as well as during a single-day event (Figure 8b, c, and d). These results were then visually represented as a difference map (Figure 8a), elucidating alterations in groundwater levels within the study area due to the boundary changes. Unsurprisingly, raising the boundary water level by 0.6 m for the duration of the simulation had widespread effects. In qualitative agreement with the machine learning model's result, the single-day event (Figure 8b) revealed a minor increase in the groundwater table in the urban area on the same day. More substantial water table elevation increases were observed at lag times ranging between 1 (Figure 8c) and 4 days (Figure 8d). The numerical model implied that a significant portion of the study area experienced

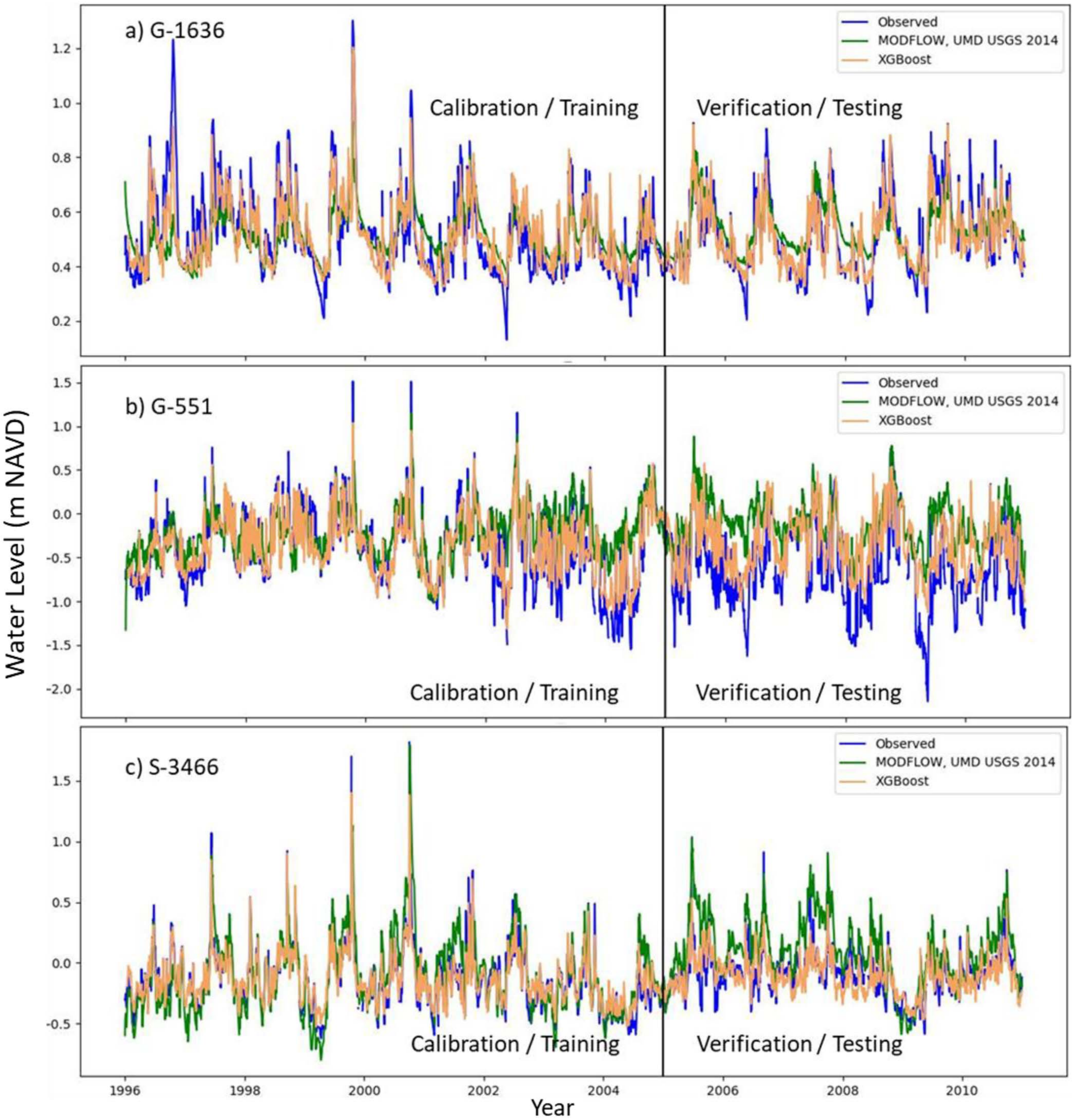

Figure 5. Examples of observed water levels (blue line) and water levels simulated with UMD model (green line) and XGBoost (orange line). The vertical line splits the calibration (training) period on the left and the verification (testing) period on the right. (a) Example of one of the smallest RMSE values for both UMD and XGBoost models. (b) Example of one of the largest RMSE values for both UMD and XGBoost models. (c) Example of groundwater site near Miami Springs wellfield where XGBoost model performance is better than that of the UMD model.

a groundwater level increase, suggesting that Everglades water levels play a pivotal role, as indicated by the XGBoost model.

An alternative approach to assessing the overall performance of the model is to utilize scatter plots encompassing all available data points across all wells. These visualizations revealed a significant level of accuracy for both models. Notably, both models exhibited superior performance during the calibration period. The UMD model achieved an impressive R^2 value of

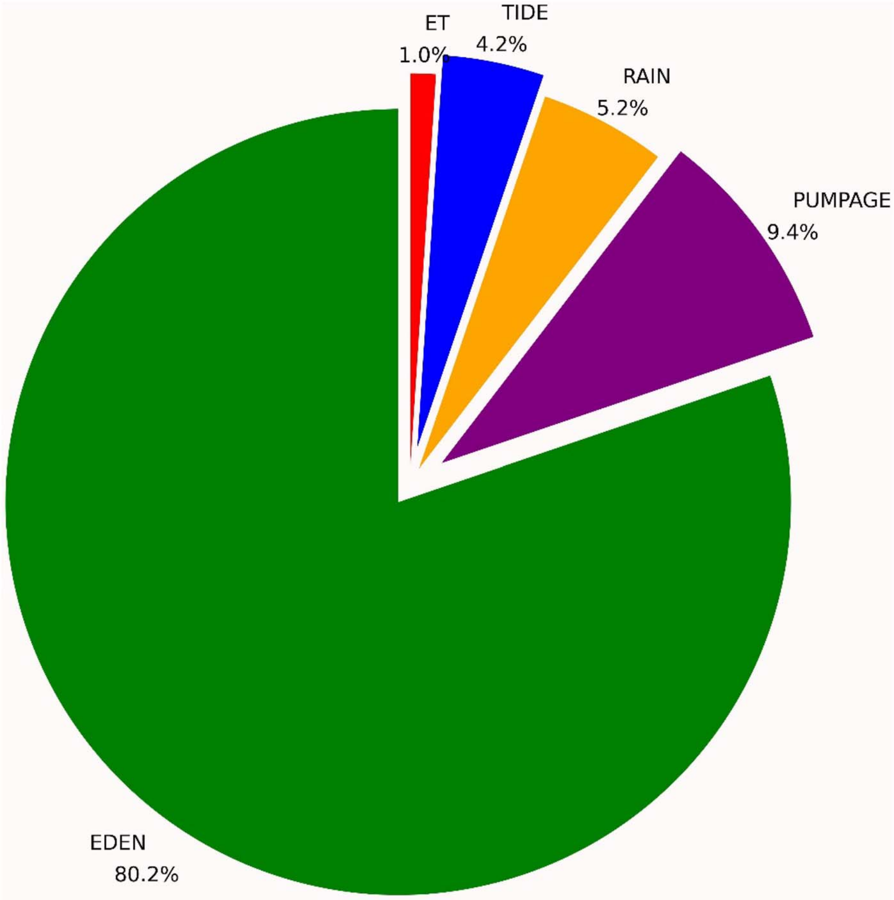

Figure 6. Distribution of wells by the most important features affecting groundwater well levels. Each slice of the pie represents a feature category: EDEN (green), ET (red), PUMPAGE (purple), RAIN (orange), and TIDE (blue).

0.92, while the XGBoost model boasted an even higher R^2 value of 0.98 (Figure 9a).

During the verification period, the UMD model maintained a commendable level of accuracy with an R^2 value of 0.90, while the XGBoost model continued to excel with an R^2 value of 0.95 (Figure 9b).

While this study focused on comparing the machine learning model with a preexisting physics-based numerical model using data up to 2010, ongoing climate change, including accelerated sea-level rise, could affect groundwater conditions beyond this period. Updating the machine learning model with more recent observations and performing forward projections would help to assess potential future impacts. Future improvements could include incorporating physically based parameters, such as distance from shore or canals and topographic elevation, into the XGBoost framework, or coupling the machine learning model with a numerical groundwater model like MODFLOW. In such a hybrid approach, the machine learning model could estimate key hydrological parameters for calibration or learn residual errors from MODFLOW simulations to adjust predictions, combining physical rigor with data-driven accuracy. Although beyond the scope of this study, these strategies represent important directions for future research and could provide more robust tools for groundwater management under changing climatic and hydrological conditions.

LIMITATIONS

The groundwater level projections generated by the XGBoost model, while valuable, do possess certain constraints. Primarily, this methodology exclusively applies to locations equipped with groundwater monitoring wells, thereby leaving spatial gaps in its coverage. Another notable limitation of the XGBoost model is its susceptibility to changes in groundwater conditions that might be brought on by changes in factors such as climate and land use, for example. Therefore, it is imperative to regularly update and maintain the model to uphold its accuracy and relevance; in some instances, however, older data that represent a different regime

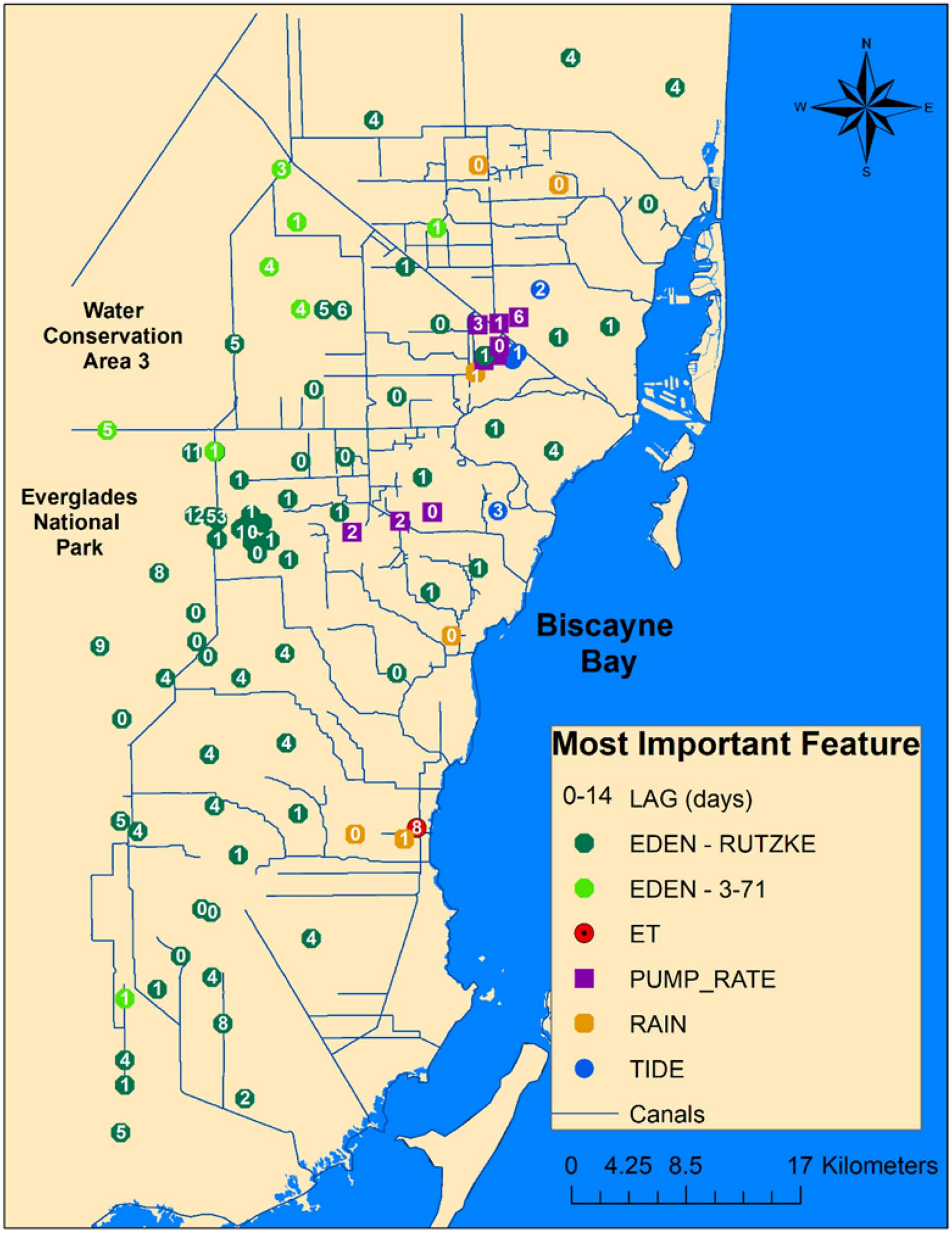

Figure 7. Map of monitoring sites showing the most important variable affecting monitoring well water levels. Numbers in shape represent the variable's lag in days. Green octagons represent two different Everglades water level stages, 3-71 stage represented by light green and Rutzke stage by dark green; red circles represent evapotranspiration; purple squares represent withdrawal well pumping rate; orange squares represent rain; and the blue circle represents the ocean water levels.

may not be particularly valuable to the model training process. Similarly, events and changes outside of the range of training data cannot be expected to be simulated with high accuracy.

The use of machine learning models such as XGBoost alongside numerical models like the UMD model underscores the potential constructive interaction between these two approaches, wherein particular modeling challenges can

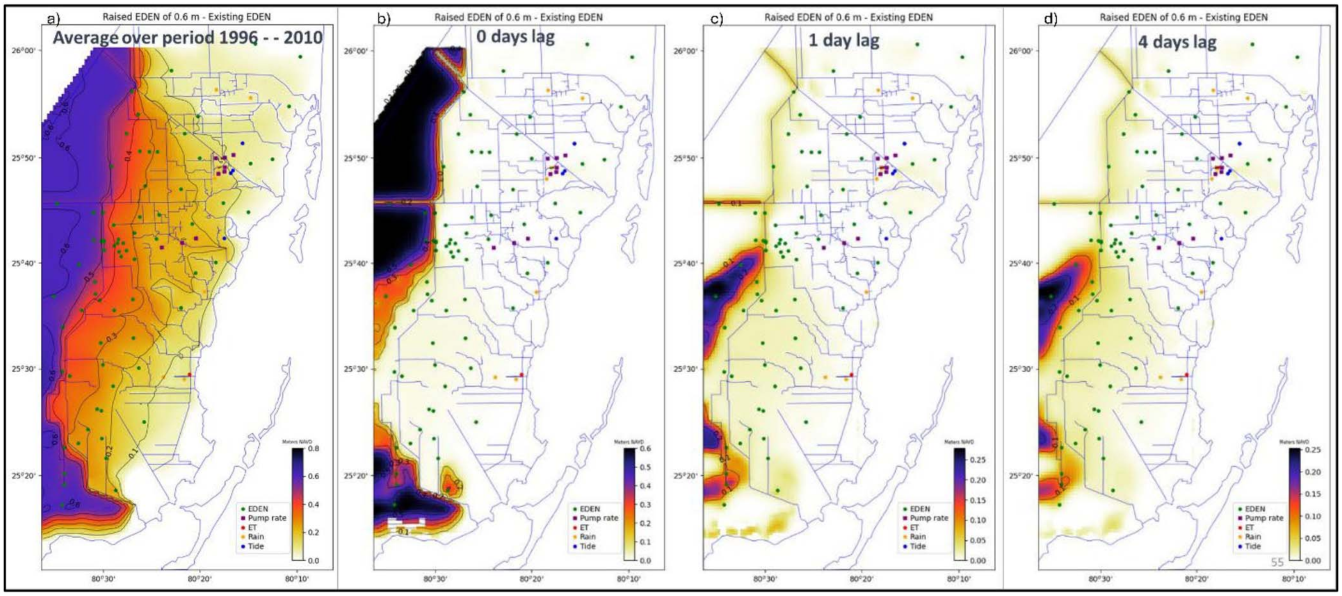

Figure 8. Maps showing differences in water table elevations between the UMD model simulation with Everglades water levels raised by 0.6 m and UMD simulation using historical Everglades water levels. Lighter shades of yellow indicate areas with slightly increased water table elevations, while darker shades of blue indicate areas with greatest increases. Four scenarios illustrated are: (a) Everglades water level raised constantly throughout simulation period (1996–2010), (b) water level change on the same day as one-day EDEN model boundary stage increase, (c) water level change 1 day after the EDEN boundary stage increase, and (d) water level change 4 days after the EDEN boundary stage increase. The map also shows monitoring sites and most influential variables affecting groundwater levels, as identified by XGBoost. Green circles represent Everglades water level stages; red circles represent evapotranspiration; purple squares represent withdrawal well pumping; orange circles represent rainfall; and blue circle represents ocean water levels.

be considered and evaluated from different perspectives. For example, the challenges faced by both models in estimating groundwater levels at groundwater monitoring sites near water supply wells showed the importance of including accurate pumping data. The monthly pumping data for Miami-Dade wells are generally considered reliable, as they undergo standard quality-assurance and quality-control procedures. However, these data are not measured directly from flow meters during operation; instead, they are estimated based on the design pump flow rates and reported operating hours.

This estimation approach can introduce errors, particularly when actual pump performance or operating conditions deviate from the design assumptions. Such discrepancies are more likely to occur in large wellfields, where small deviations across multiple wells can accumulate. Consequently, these potential inaccuracies in the pumping data could contribute to the higher RMSE values observed in the vicinity of large wellfields.

Despite the constraints inherent in this modeling approach, it offers the advantage of explicitly identifying the most important drivers of groundwater behavior for specific sites.

SUMMARY

This study investigated the use of the machine learning technique XGBoost to predict groundwater levels in Miami-Dade County and compared the results with those of an existing calibrated numerical model, the USGS UMD model (Hughes and White, 2016).

An alternative approach to assessing the overall performance of the model is to utilize scatter plots encompassing all available data points across all wells. These visualizations revealed a significant level of accuracy for both models. Notably, both models exhibited superior performance during the calibration period. The UMD model achieved an impressive R^2 value of 0.97, while the XGBoost model boasted an even higher R^2 value of 0.99 (Figure 9a).

The XGBoost model gave better overall results than the UMD model. The lowest accuracy in the XGBoost model was in the monitoring wells with large amounts of missing data and extremely short time series. Errors in pumping rate data could cause high RMSE near municipal wells for both models; however, XGBoost can better overcome these errors. There are groundwater sites near municipal wells where the XGBoost model performed well but the UMD model did not.

XGBoost could be used in future studies to improve prediction performance in problematic areas, such as areas with high withdrawal rates. Most groundwater site water levels are correlated primarily with water levels in the Everglades at various time lags and then with production well withdrawal rates, rainfall amounts, and ocean

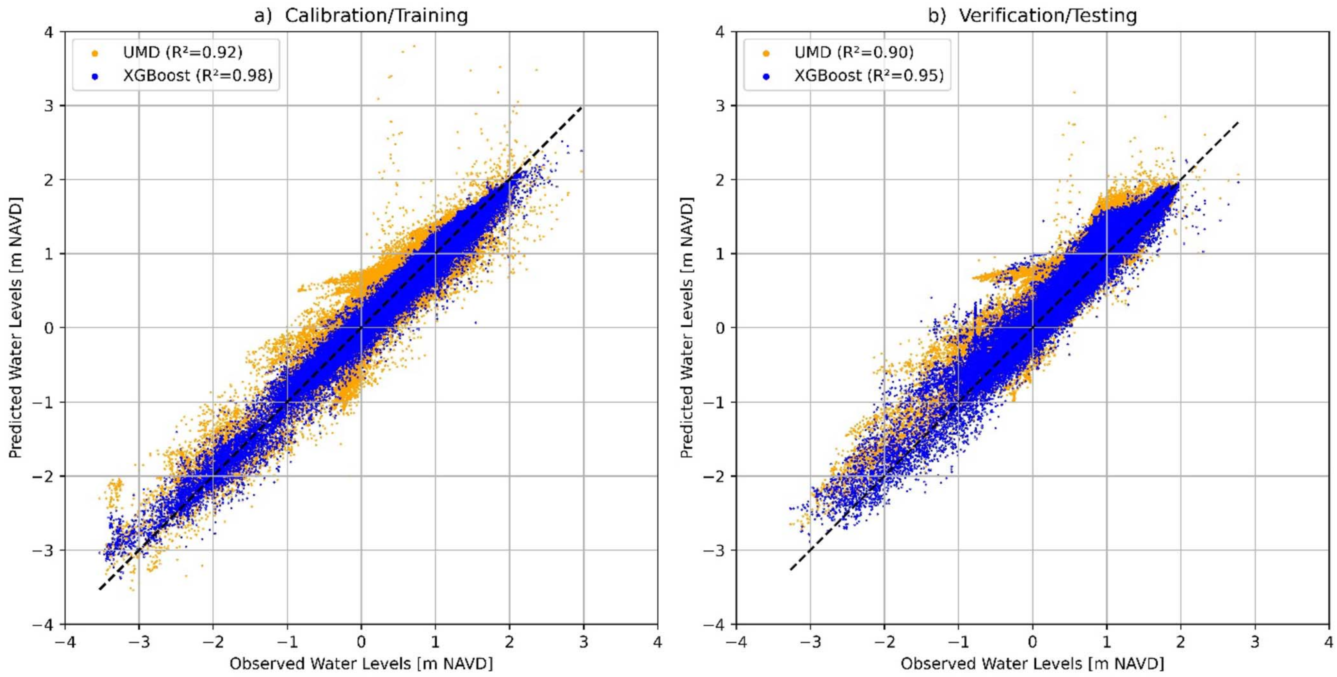

Figure 9. Scatter plots encompassing all available data points across all wells for UMD model represented by orange markers and XGBoost represented by blue markers for: (a) calibration period and (b) verification period.

water levels. The XGBoost model created in this research had a singular focus on groundwater level time-series analysis, without considering spatial relations between wells, mass balance, and canal-aquifer interactions.

The findings of this research might be especially beneficial for understanding how the restoration of Everglades water levels could affect the groundwater table in the urban area, though drainage and the associated active water table management were not considered in this machine learning model. The model could likely be improved by incorporating daily withdrawal rate data for all wellfield sites, including non-MDWASD wellfield sites. The XGBoost model could be used in hourly time steps to get a better understanding of the effect of rainfall and high tide events on groundwater levels.

SUPPLEMENTAL MATERIAL

Supplemental Material associated with this article can be found online at https://doi.org/10.2113/EEG-D-25-00016.

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